

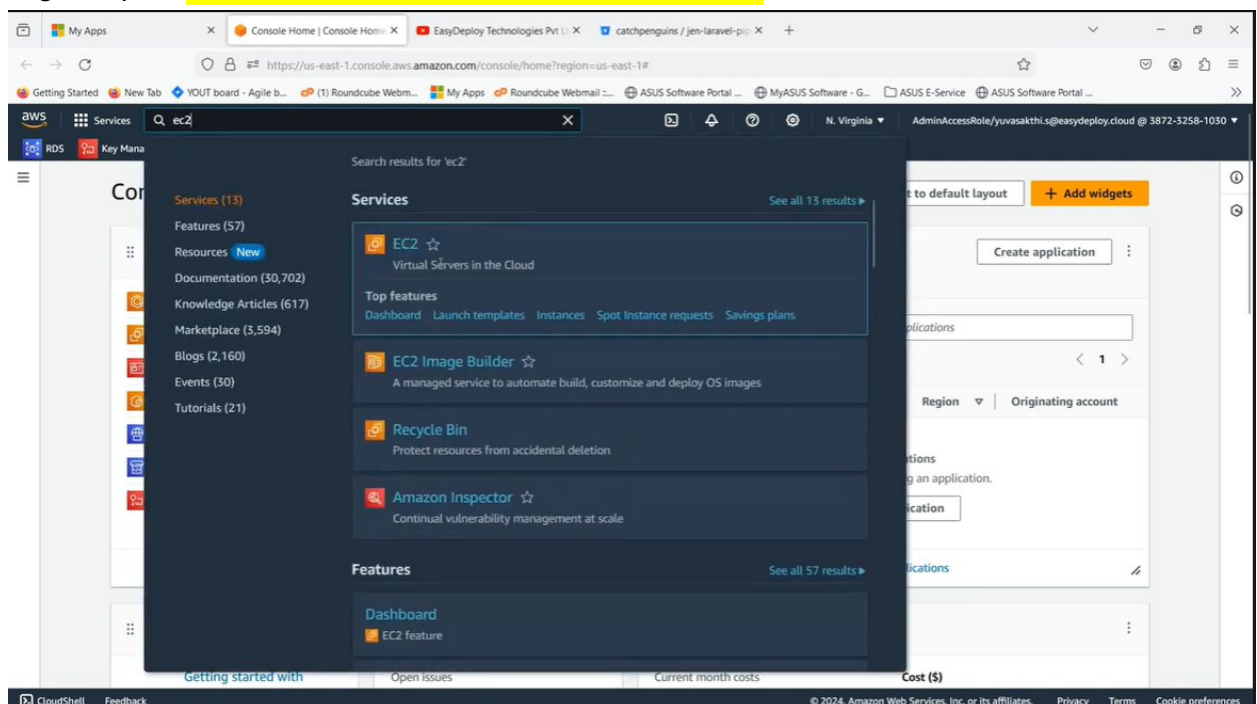
How to set up Jenkins CI CD pipeline to automate the build and deployment of Laravel to AWS ECS

First, We Study about the Jenkins. What is Jenkins?

- Jenkins is an open-source automation server used primarily for continuous integration and continuous delivery (CI/CD) Pipelines.
- It allows developers to automate the process of building, testing and deploying software applications.
- Here how it works:
 - 1) Developers commit code changes to a version control system like GIT.
 - 2) Jenkins monitors the version control system for changes
 - 3) When changes are detected, Jenkins pulls the code, initiates the build process, and runs automated tests.
 - 4) If the build and tests are successful, Jenkins can deploy the application to various environments, such as development, staging, or production.

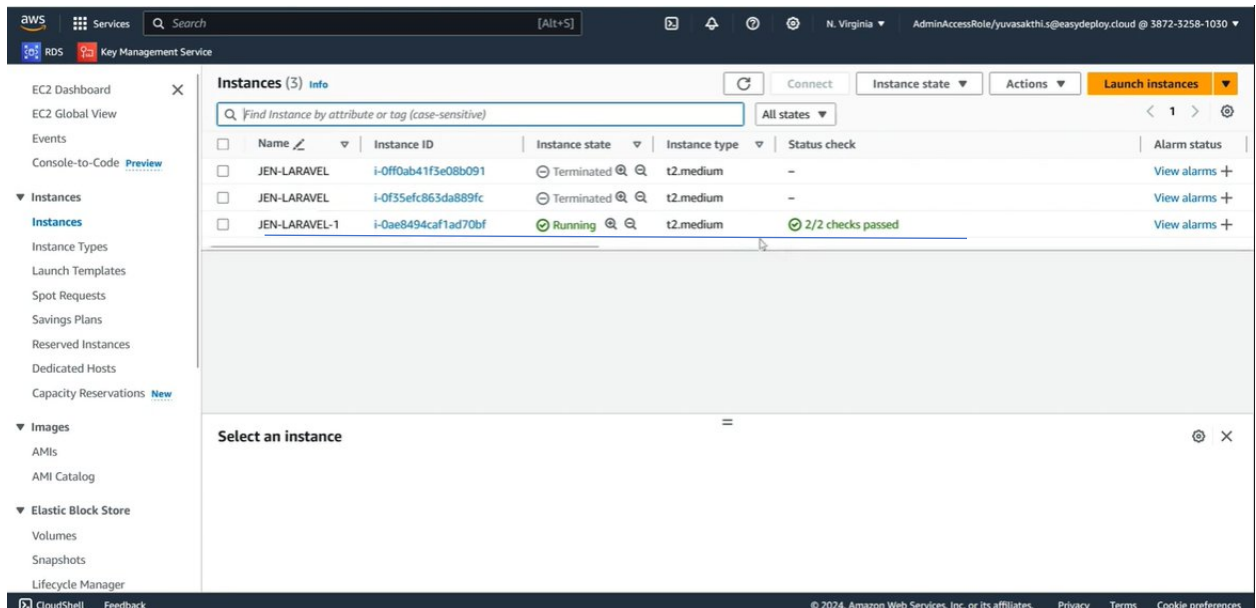
Now, Moving towards the steps:

- 1) Login to your **AWS account----->Dashboard---->Search EC2**



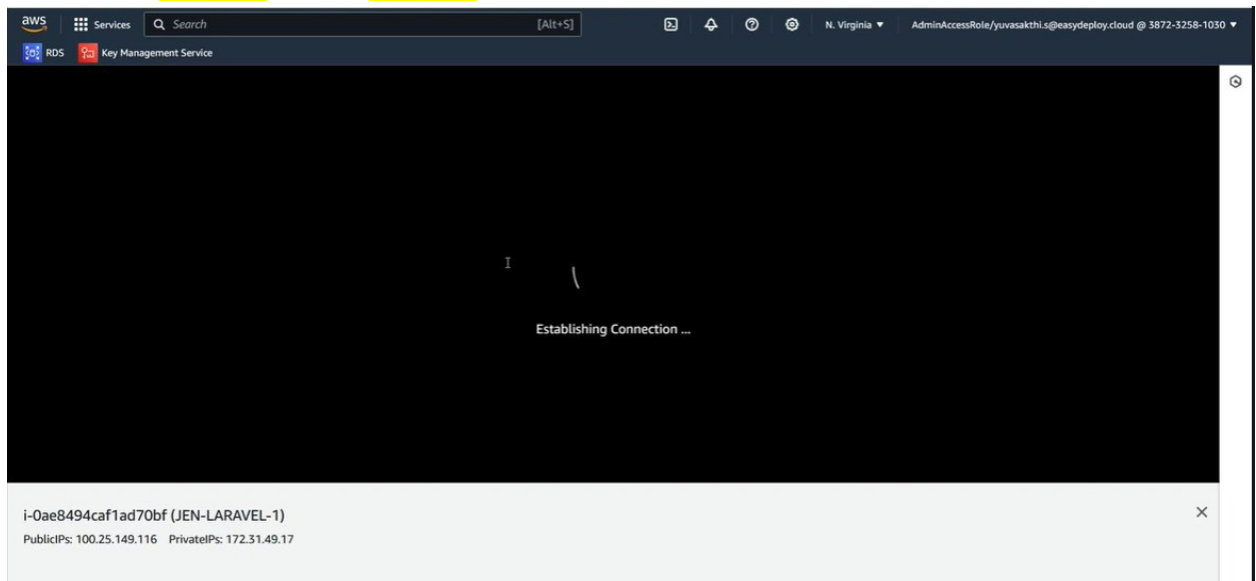
- 2)

3) Create EC2 Instance with instance type “t2.medium”



4) Click on the running Jenkins instance-à scroll down to Security groups à Open the port “8080”, “22”, “80”, “443”

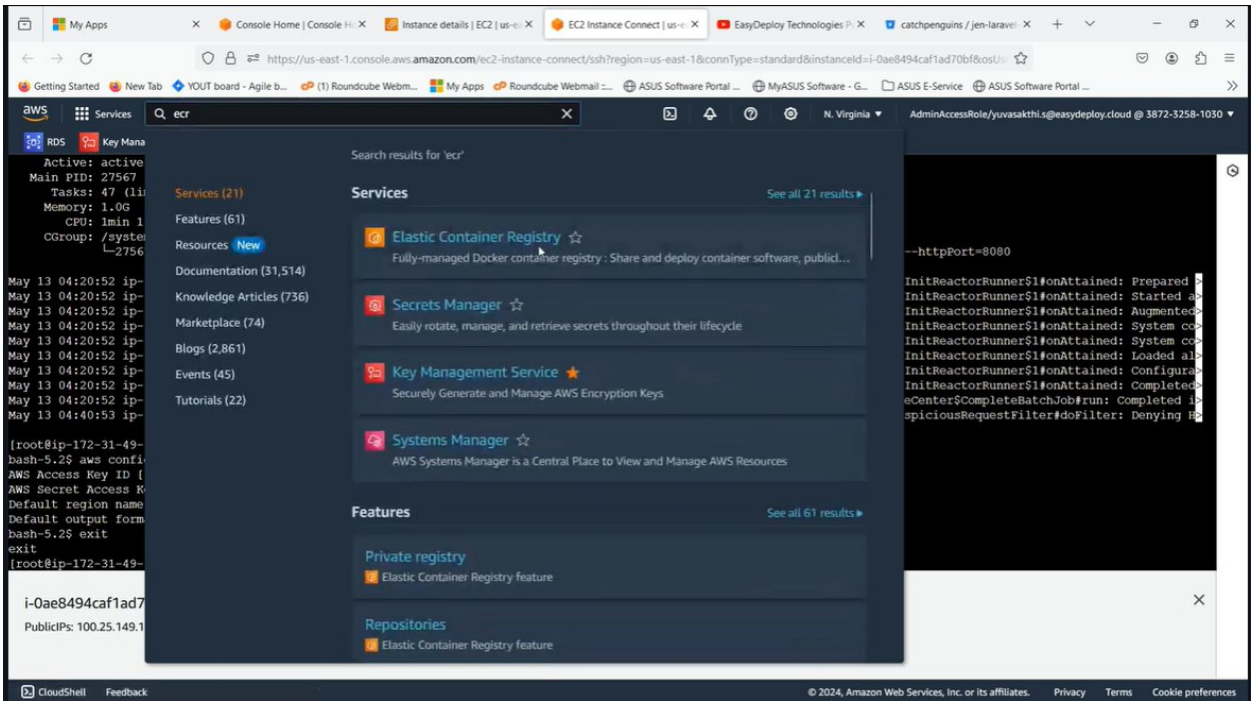
5) Now, click “Connect” then “Connect”



6) Type: `sudo su` to change user to root user

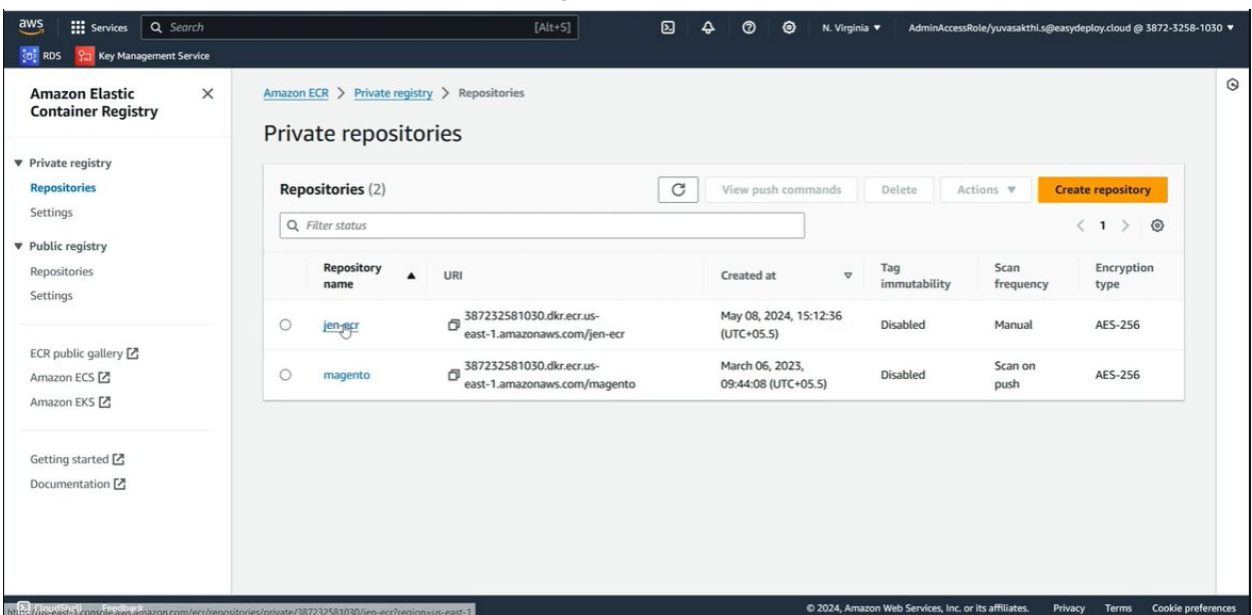
7) `systemctl status jenkins.service`

8) `sudo su -s/bin/bash Jenkins bash-5.2s aws configure`

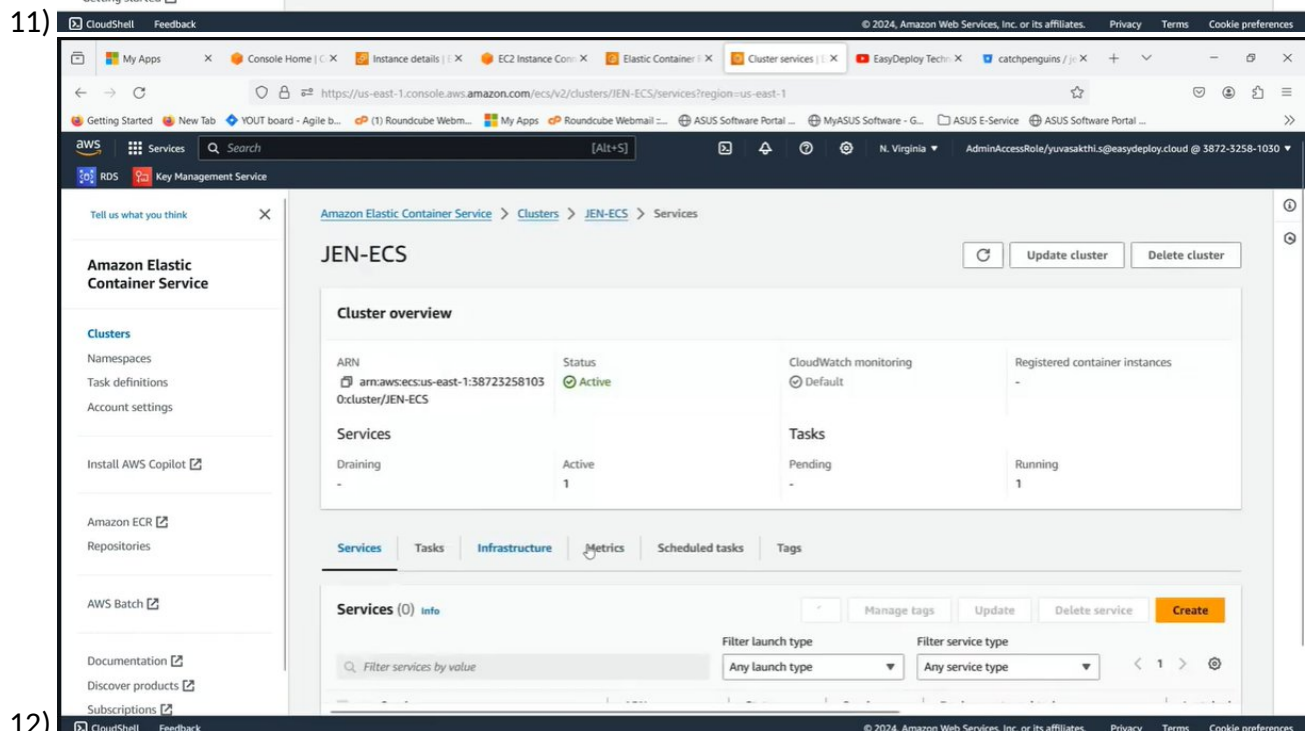
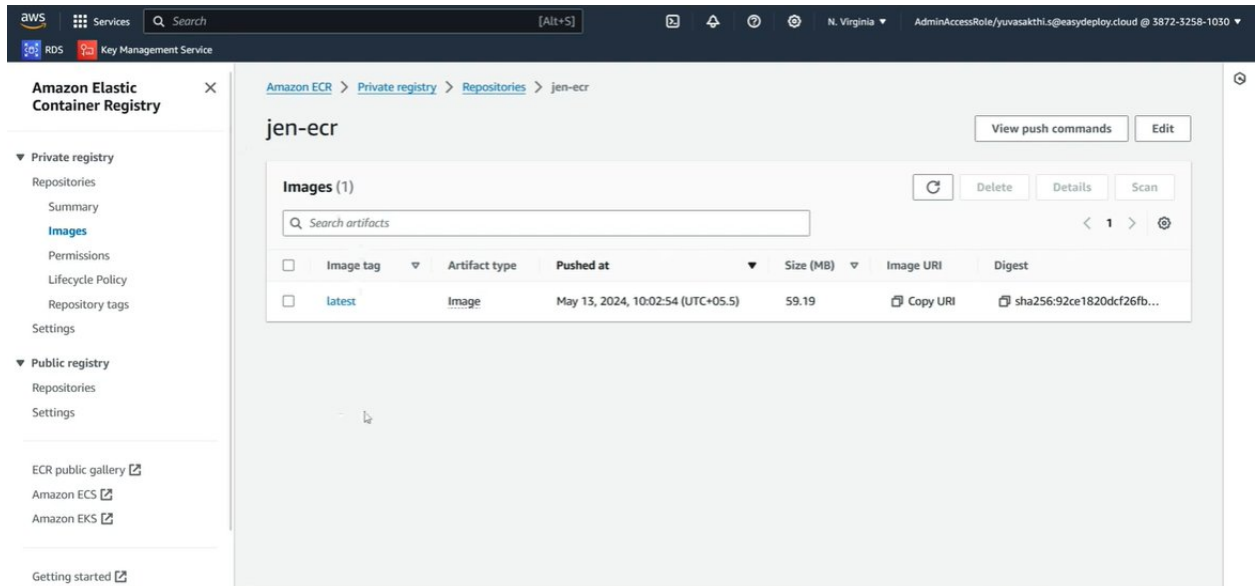


9)

Now Type and search “Elastic container Registry” | “ECR”



10)



- 12)
- 13) Now go to "ECS" as shown above
- 14) Now go to instance you create "EC2 i.e Jenkins Instance"
- 15) Copy Public Url followed by colon 8080 like this **ex.10.0.149.116:8080**
- 16) Login to Jenkins admin,admin

17)

The screenshot shows a web browser window with the Jenkins login page. The browser's address bar shows the URL `100.25.149.116:8080/login?from=%2F`. The login page features the Jenkins logo (a cartoon man in a tuxedo) on the left and a "Sign in to Jenkins" form on the right. The form includes fields for "Username" (containing "admin") and "Password" (masked with dots), a "Keep me signed in" checkbox, and a blue "Sign in" button. Below the login page, the Jenkins dashboard is visible. The dashboard has a dark header with the Jenkins logo, a search bar, and a "log out" link. The main content area is divided into two columns. The left column contains a sidebar with links: "New Item", "People", "Build History", "Manage Jenkins", and "My Views". Below these links are two expandable sections: "Build Queue" (showing "No builds in the queue.") and "Build Executor Status" (showing two "idle" executors). The right column contains a "Welcome to Jenkins!" message, a "Start building your software project" section with a "Create a job" button, and a "Set up a distributed build" section with buttons for "Set up an agent", "Configure a cloud", and "Learn more about distributed builds".

Sign in to Jenkins

Username
admin

Password

☐ Keep me signed in

Sign in

Jenkins

Dashboard

+ New Item

People

Build History

Manage Jenkins

My Views

Build Queue
No builds in the queue.

Build Executor Status
1 idle
2 idle

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API Jenkins 2.440.3

18) Click on "New Item" -> Manage Jenkins -> Install Plugins

19)

Jenkins Search (CTRL+K) admin nga log out

Dashboard > Manage Jenkins > Plugins

Plugins Install

Updates
Available plugins
Installed plugins
Advanced settings
Download progress

Install	Name	Released
✓	Amazon Web Services SDK : ECS 1.12.696-451.v0651a_da_9ca_ec Library plugins (for use by other plugins) aws ECS module for the AWS SDK for Java.	1 mo 3 days ago
✓	Amazon Elastic Container Service (ECS) / Fargate 1.49 Cluster Management Agent Management aws Use Amazon EC2 Container Service to provide elastic agents.	3 mo 16 days ago
✓	AWS Credentials 231.v08a_59f17d742 aws Allows storing Amazon IAM credentials within the Jenkins Credentials API. Store Amazon IAM access keys (AWSAccessKeyId and AWSSecretKey) within the Jenkins Credentials API. Also support IAM Roles and IAM MFA Token.	1 mo 15 days ago
✓	Amazon Web Services SDK : ECR 1.12.696-451.v0651a_da_9ca_ec Library plugins (for use by other plugins) aws ECR module for the AWS SDK for Java.	1 mo 3 days ago
✓	Amazon ECR 1.114.vfd22430621f5 aws This plugin generates Docker authentication token from Amazon Credentials to access Amazon ECR.	1 yr 3 mo ago

20)

Bitbucket Your work Pull requests Repositories Projects People More Create Search

JEN-LARAVEL-PIPELINE EasyDeploy / CICD Invite Clone

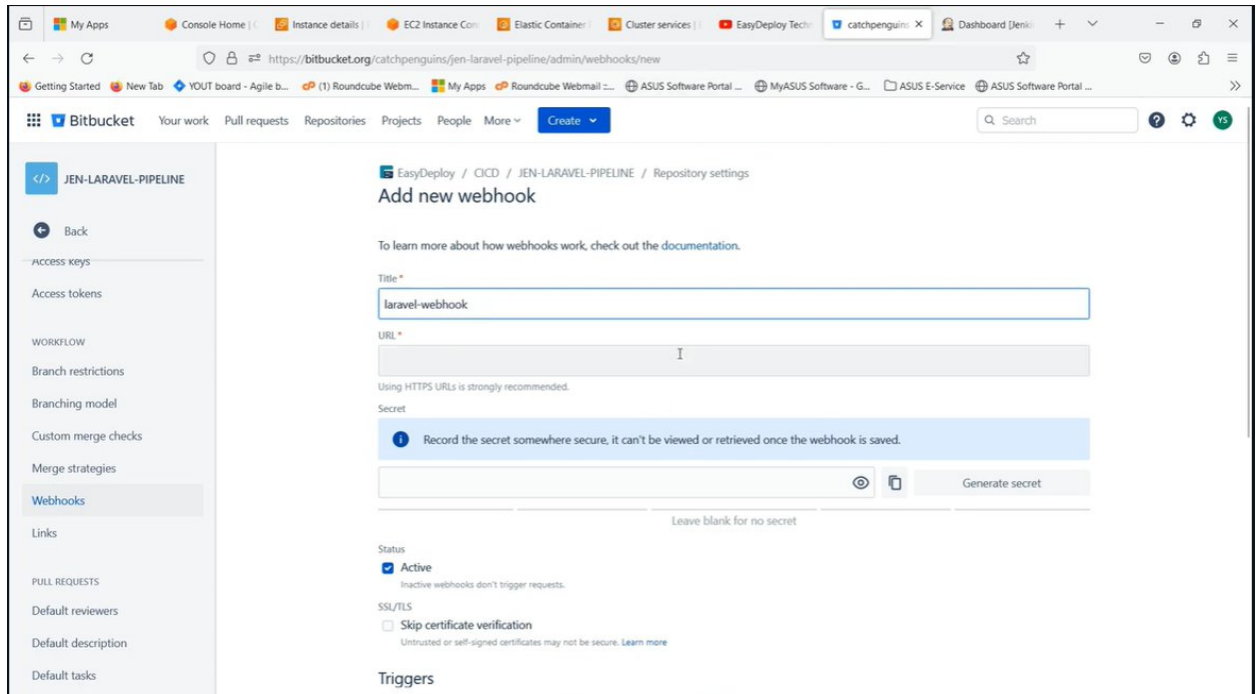
Here's where you'll find this repository's source files. To give your users an idea of what they'll find here, add a description to your repository.

Name	Size	Last commit	Message
app		3 days ago	committed
bootstrap		3 days ago	committed
config		3 days ago	committed
database		3 days ago	committed
public		3 days ago	committed
resources		3 days ago	welcome.blade.php edited online with Bitbucket
routes		3 days ago	committed
storage		3 days ago	committed
tests		3 days ago	committed
.editorconfig	258 B	3 days ago	first commit

Repository details
Last updated 17 minutes ago
Open pull requests 0
Branches 1
Watchers 2
Forks 0
Access level Admin

0 builds
It looks like you haven't configured a build tool yet. You can use Bitbucket Pipelines to build, test and deploy your code.
Your existing plan already includes build minutes.
[Set up a pipeline](#)

Visit to "Bitbucket" where the code is present -> Repository settings -> Webhooks

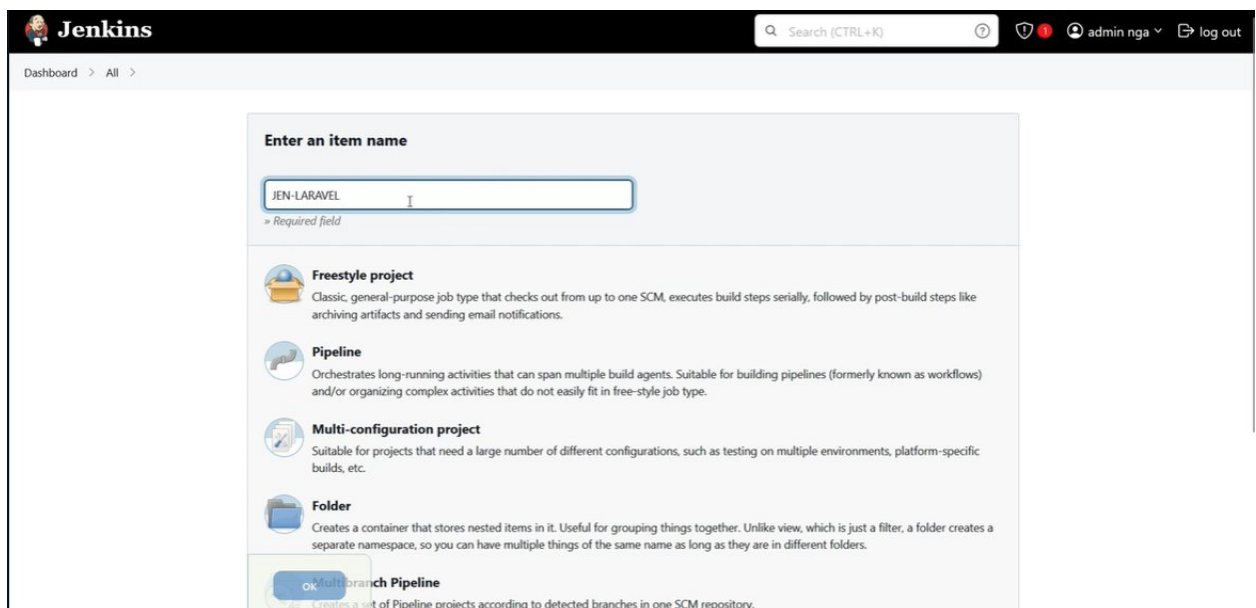


Copy the Jenkins url and paste into the url field “Bitbucket”



Click

on Generate Secret Scroll down to “Save”, Now goto Jenkins->create new item->as shown in image below



Click on “Pipeline” --> Enable Github Project à go to “Bitbucket” clone the repository url copy and paste into

The screenshot shows the Jenkins configuration interface for a project named "JEN-LARAVEL". The left sidebar contains a "Configure" section with sub-items: "General", "Advanced Project Options", and "Pipeline". The "General" tab is currently selected.

In the "General" tab, the following options are visible:

- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☒ Build when a change is pushed to BitBucket
- Override Repository URL or Path:
- ☐ Build on branch created
- ☐ GitHub hook trigger for GITScm polling ?
- ☐ Poll SCM ?
- ☐ Quiet period ?
- ☐ Trigger builds remotely (e.g., from scripts) ?

Below these options is a section titled "Advanced Project Options" with a dropdown menu set to "Advanced".

The "Pipeline" section is partially visible at the bottom of the configuration page, showing:

- ☐ GitHub hook trigger for GITScm polling ?
- ☒ Poll SCM ?
- Schedule ? :
- No schedules so will only run due to SCM changes if triggered by a post-commit hook
- ☐ Ignore post-commit hooks ?
- ☐ Quiet period ?
- ☐ Trigger builds remotely (e.g., from scripts) ?

At the bottom of the configuration page, there are "Save" and "Apply" buttons.

Configure

General

Advanced Project Options

Pipeline

Pipeline

Definition

Pipeline script from SCM

SCM ?

Git ?

Repositories ?

Repository URL ?

Credentials ?

+ Add

Jenkins

Save

Apply

Repository URL ?

https://yuvasakthi07@bitbucket.org/catchpenguins/jen-laravel-pipeline.git

❗ Failed to connect to repository : Error performing git command: git ls-remote -h https://yuvasakthi07@bitbucket.org/catchpenguins/jen-laravel-pipeline.git HEAD

Jenkins Credentials Provider: Jenkins

Add Credentials

Domain

Global credentials (unrestricted)

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

☐ Treat username as secret ?

Password ?

Jenkins Credentials Provider: Jenkins

Global (Jenkins, nodes, items, all child items, etc)

Username ?

yuvasakthi S

☐ Treat username as secret ?

Password ?

.....

ID ?

bitbucket

Description ?

Cancel

Add

My Apps Console Home Instance details EC2 Instance X Elastic Container Cluster services EasyDeploy Tech catchpenguins JEN-LARAVEL Co

Getting Started New Tab YOUT board - Agile b... (1) Roundcube Webm... My Apps Roundcube Webmail... ASUS Software Portal... MyASUS Software - G... ASUS E-Service ASUS Software Portal...

aws Services Q ecr N. Virginia AdminAccessRole/yuvaskthls@easydeploy.cloud @ 3872-3258-1030

RDS Key Management Service

Active: active (running) since Mon 2024-05-13 04:14:33 UTC; 45min ago
Main PID: 27567 (java)
Tasks: 47 (limit: 4659)
Memory: 1.0G
CPU: lmin 16.581s
CGroup: /system.slice/jenkins.service
└─27567 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

```
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.164+0000 [id=260] INFO jenkins.InitReactorRunner$1#onAttained: Prepared
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.193+0000 [id=259] INFO jenkins.InitReactorRunner$1#onAttained: Started
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.195+0000 [id=259] INFO jenkins.InitReactorRunner$1#onAttained: Augmented
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.223+0000 [id=262] INFO jenkins.InitReactorRunner$1#onAttained: System co
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.223+0000 [id=262] INFO jenkins.InitReactorRunner$1#onAttained: System co
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.228+0000 [id=261] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.236+0000 [id=262] INFO jenkins.InitReactorRunner$1#onAttained: Configur
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.723+0000 [id=261] INFO jenkins.InitReactorRunner$1#onAttained: Complet
May 13 04:20:52 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:20:52.725+0000 [id=74] INFO h.m.UpdateCenter$CompleteBatchJob#run: Complet
May 13 04:40:53 ip-172-31-49-17.ec2.internal jenkins[27567]: 2024-05-13 04:40:53.111+0000 [id=17] WARNING j.s.SuspiciousRequestFilter#doFilter: Denying
```

```
[root@ip-172-31-49-17 ec2-user]# sudo su -s /bin/bash jenkins
bash-5.2$ aws configure
AWS Access Key ID [*****E0BS]:
AWS Secret Access Key [*****WJS2]:
Default region name [us-east-1]:
Default output format [None]:
bash-5.2$ exit
exit
[root@ip-172-31-49-17 ec2-user]# yum install
```

i-Oae8494caf1ad70bf (JEN-LARAVEL-1)
PublicIPs: 100.25.149.116 PrivateIPs: 172.31.49.17

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In Jenkins-> branch specifier /main, script path JenkinsFile -> Build Now

aws Services Q ecr N. Virginia AdminAccessRole/yuvaskthls@easydeploy.cloud @ 3872-3258-1030

RDS Key Management Service

Transaction test succeeded.
Running transaction

```
Preparing : 1/1
Installing : git-core-2.40.1-1.amzn2023.0.2.x86_64 1/8
Installing : git-core-doc-2.40.1-1.amzn2023.0.2.noarch 2/8
Installing : perl-lib-0.65-477.amzn2023.0.6.x86_64 3/8
Installing : perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64 4/8
Installing : perl-File-Find-1.37-477.amzn2023.0.6.noarch 5/8
Installing : perl-Error-1:0.17029-5.amzn2023.0.2.noarch 6/8
Installing : perl-Git-2.40.1-1.amzn2023.0.2.noarch 7/8
Installing : git-2.40.1-1.amzn2023.0.2.x86_64 8/8
Running scriptlet: git-2.40.1-1.amzn2023.0.2.x86_64 8/8
Verifying : git-2.40.1-1.amzn2023.0.2.x86_64 1/8
Verifying : git-core-2.40.1-1.amzn2023.0.2.x86_64 2/8
Verifying : git-core-doc-2.40.1-1.amzn2023.0.2.noarch 3/8
Verifying : perl-Error-1:0.17029-5.amzn2023.0.2.noarch 4/8
Verifying : perl-File-Find-1.37-477.amzn2023.0.6.noarch 5/8
Verifying : perl-Git-2.40.1-1.amzn2023.0.2.noarch 6/8
Verifying : perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64 7/8
Verifying : perl-lib-0.65-477.amzn2023.0.6.x86_64 8/8
```

```
Installed:
git-2.40.1-1.amzn2023.0.2.x86_64 git-core-2.40.1-1.amzn2023.0.2.x86_64 git-core-doc-2.40.1-1.amzn2023.0.2.noarch
perl-Error-1:0.17029-5.amzn2023.0.2.noarch perl-File-Find-1.37-477.amzn2023.0.6.noarch perl-Git-2.40.1-1.amzn2023.0.2.noarch
perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64 perl-lib-0.65-477.amzn2023.0.6.x86_64
Complete!
```

```
[root@ip-172-31-49-17 ec2-user]# sudo chmod 666 /var/run/docker.sock
```

Task overview

ARN	Last status	Desired status	Started/Created at
arn:aws:ecs:us-east-1:38723258103:task/JEN-ECS/e32937cc52df43a997d0db2d0ebbf691	Running	Running	May 13, 2024 at 10:45 (UTC+5:30) May 13, 2024 at 10:44 (UTC+5:30)

Configuration

Operating system/Architecture	Capacity provider	ENI ID	Public IP
Linux/X86_64	-	eni-08281fbc45bd9344	34.224.2.183 open address
CPU Memory .5 vCPU 1 GB	Launch type FARGATE	Network mode awsvpc	Private IP 172.31.59.75
Platform version 1.4.0	Container instance ID -	Subnet ID subnet-918af6af	MAC address 06:07:51:bc:34:0f
Task definition: revision TASKDEFINITION-LARAVEL:7			
Task group service:JEN-SERVICE			

Container details for JEN-ECR

Copy the public, paste and hit enter , you will see the Laravel screen

WELCOME TO X

107.23.122.72

Getting Started New Tab YOUT board - Agile b... (1) Roundcube Webm... My Apps Roundcube Webmail ... ASUS Software Portal ... MyASUS Software - G... ASUS E-Service ASUS Software Portal ...



Documentation

Laravel has wonderful, thorough documentation covering every aspect of the framework. Whether you are new to the framework or have previous experience with Laravel, we recommend reading all of the documentation from beginning to end.

Laracasts

Laracasts offers thousands of video tutorials on Laravel, PHP, and JavaScript development. Check them out, see for yourself, and massively level up your development skills in the process.

Laravel News

Laravel News is a community driven portal and newsletter aggregating all of the latest and most important news in the Laravel ecosystem, including new package releases and tutorials.

Vibrant Ecosystem

Laravel's robust library of first-party tools and libraries, such as [Forge](#), [Valet](#), [Nova](#), and [Envoyer](#) help you take your projects to the next level. Pair them with powerful open source libraries like [Cashier](#), [Dusk](#), [Echo](#), [Horizon](#), [Sanctum](#), [Telescope](#), and more.

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Laravel v8.83.27 (PHP v8.2.19)